

BÁO CÁO KIỂM ĐỊNH CUỐI DATASET E_i

30 core audit rounds + 10 model-sanity rounds - SOURCE FINAL v5.0.0

Phạm vi. Ba lớp kiểm định cốt lõi gồm: 10 vòng đối chiếu nguồn, 10 vòng integrity của file dataset train, và 10 vòng final quality gate. Ngoài ra có thêm 10 vòng model-sanity trên các positive feature profiles chưa thấy, không có train/test profile overlap.

A. 10 vòng đối chiếu cơ sở tạo dataset

Round	Check	Status	Sources	Action/Boundary
1	Authority and intended scope	PASS_WITH_BOUNDARY	S1	Do not claim the app implements WHO triage; use WHO-informed/guideline-derived wording.
2	Binary target semantics	PASS_WITH_BOUNDARY	S1;S2	Target renamed/interpreted narrowly: presence of at least one of five selected mapped RED signs, not the full IIT RED category.
3	unresponsive mapping	PASS	S2	Keep canonical feature name unresponsive.
4	respiratory distress / cyanosis mapping	PASS_WITH_BOUNDARY	S2;S7	Rename feature to respiratory_distress_or_cyanosis; UI may simplify wording but report preserves canonical source term.
5	heavy bleeding mapping	PASS_WITH_BOUNDARY	S2	Keep heavy_bleeding; disclose citizen-observed/group-level aggregation.
6	active convulsions mapping	PASS_AFTER_CHALLENGE	S2	Replace training feature seizure with active_convulsions; app alias seizure is allowed only when it means active current convulsions.

7	high-risk trauma mapping	PASS_AFTER_CHA_NGE	S2;S7	Replace training feature major_trauma with high_risk_trauma and require UI helper examples.
8	mobility / cannot_move	EXCLUDED_FROM_LR	S3;S4;S5	Remove cannot_move from LR training features; retain only as structured context in the app/source registry.
9	Patient-level source vs group-report aggregation	ADAPTATION_DISCLOSED	S1;S2;S3	Master CSV explicitly states group-level OR semantics: a flag is 1 if at least one person in the report has the sign. This is project adaptation, not source-validated.
10	Anti-AI-generation provenance	PASS	S2	Final training rows are generated by deterministic exhaustive 2^5 enumeration; no LLM, random sampling or free-form scenario generation.

Kết quả nguồn không bị ép thành PASS tuyệt đối. Các mục PASS_WITH_BOUNDARY, EXCLUDED_FROM_LR và ADAPTATION_DISCLOSED được giữ lại vì chính các giới hạn này giúp tránh claim vượt quá nguồn.

B. 10 vòng integrity audit của dataset train

Round	Check	Status	Detail
1	Schema and required columns	PASS	Columns=['profile_id', 'unresponsive', 'respiratory_distress_or_cyanosis', 'heavy_bleeding', 'active_convulsions', 'high_risk_trauma', 'urgency_label']
2	Missing values and binary domains	PASS	missing_rows=0; binary_domains=True
3	Row count equals	PASS	rows=32 expected=32

	complete 2 ⁵ state space		
4	Unique feature profiles / duplicate audit	PASS	unique_profiles=32/32
5	Exhaustive feature-profile coverage	PASS	All 32 Boolean combinations present exactly once.
6	Label-rule consistency	PASS	urgency_label = OR of five selected source-mapped RED signs for every row.
7	Target leakage / irrelevant-feature exclusion	PASS	Excluded from train CSV: ['cannot_move', 'description', 'injured_count', 'reference_triage_class', 'scenario_text_vi', 'severe_condition']
8	Master provenance + label basis + source IDs	PASS	master_complete=True; source_trace=True
9	Deterministic rebuild equivalence	PASS	Builder logic reproduces the same ordered 32 protocol states and labels.
10	Class imbalance is explicit, not hidden duplicate balancing	PASS	class_counts={0: 1, 1: 31}; training script uses class_weight='balanced' instead of duplicating negative rows.

C. 10 vòng final quality gate toàn package

Round	Check	Status	Detail
1	Source registry resolution	PASS	7 verified reference records with explicit authority, role and URL.
2	Canonical feature/source alignment	PASS	Model feature order=['unresponsive', 'respiratory_distress_or_cyanosis', 'heavy_bleeding', 'active_convulsions', 'high_risk_trauma'].
3	Old/ambiguous train features removed	PASS	seizure, major_trauma, cannot_move, injured_count and severe_condition are absent from final train CSV.
4	Master provenance/label basis/source IDs	PASS	Every master row contains label_basis,

	completeness		source_ids, generation_method, builder hash and explicit non-LLM/non- clinical flags.
5	Deterministic rebuild equivalence	PASS	Independent reconstruction matches all 32 ordered profiles and labels.
6	Training script executable	PASS	Return code=0; model artifact regenerated successfully=True.
7	Coefficient sign and symmetry sanity	PASS	All 5 coefficients positive; max- min=0.000e+00.
8	E_i monotonicity and threshold separation	PASS	80 one-bit 0->1 transitions checked; min delta=0.021878; negative=0.257683; min positive=0.512681.
9	DOCX render and visual QA	PASS	Both final DOCX reports rendered to PDF/PNG and were visually inspected for clipping/overlap.
10	Archive integrity preflight	PASS	Temporary full-package ZIP passed CRC/integrity test before final packaging.

D. Phụ lục - 10 vòng model sanity trên unseen positive profiles

State space final có đúng một all-zero negative profile. Vì không thể đặt cùng negative profile ở cả train và test mà vẫn bảo đảm profile overlap = 0, sanity test giữ negative profile trong train và chỉ hold-out các positive profiles. Metric chính là recall trên các tổ hợp positive chưa thấy; đây không phải clinical evaluation.

Round	Train unique	Held-out +	Overlap	Recall	Mean E_i
1	28	4	0	0.750	0.809
2	29	3	0	0.667	0.701
3	29	3	0	1.000	0.864
4	29	3	0	0.667	0.701
5	29	3	0	1.000	0.864
6	29	3	0	0.667	0.701
7	29	3	0	1.000	0.864
8	29	3	0	0.667	0.701
9	29	3	0	1.000	0.864
10	29	3	0	1.000	0.816

Mean unseen-positive recall = 0.8417. Đây là technical sanity metric בלבד, không phải clinical accuracy.

E. Thay đổi dataset sau source audit

- Loại cannot_move khỏi Logistic Regression; vẫn giữ như app context vì mobility có cơ sở nhưng không phải selected RED-sign target.
- Đổi seizure -> active_convulsions để khớp thuật ngữ WHO IITT và tránh hiểu nhầm tiền sử co giật.
- Đổi major_trauma -> high_risk_trauma để khớp WHO IITT/reference card.
- Đổi respiratory_distress -> respiratory_distress_or_cyanosis để giữ đủ canonical source criterion.
- Giảm state space từ $2^6=64$ xuống $2^5=32$ unique profiles; không duplicate.
- Bỏ cân bằng bằng duplicate/sample-weight trong dataset; class balancing nằm trong training script qua class_weight='balanced'.

F. Model artifact và rule baseline

Final LR fit trên toàn bộ 32 protocol states đạt full-state fidelity = 1.0000; negative E_i = 0.2577; minimum positive E_i = 0.5127. Direct OR rule vẫn là baseline chính xác của chính definition target. Logistic Regression chỉ là learned lightweight surrogate; không được claim superior về mặt y khoa.

G. Claim boundary bắt buộc

- Không gọi đây là 'WHO dataset' hoặc nói WHO xác nhận 32 dòng.
- Không gọi label là expert-confirmed clinical ground truth.
- Không nói ứng dụng implements full WHO/IITT triage.
- Không gọi urgency_label=0 là 'safe' hay 'non-urgent' theo đầy đủ IITT.
- Không gọi E_i là mortality probability, clinical probability hay dispatch probability.
- Phải công bố group-level OR aggregation là project adaptation.

H. Nguồn cơ sở

[1] Interagency Integrated Triage Tool (IITT) overview. World Health Organization.

<https://www.who.int/tools/triage>

[2] Interagency Integrated Triage Tool: Age ≥ 12 . WHO / ICRC / MSF.

https://apps.who.int/gb/MSPI/pdf_files/2022/10/Item1_27-10.pdf

[3] Facility-Based Mass Casualty Triage - Guidance Note (MC-IITT). World Health Organization.

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[4] START Adult Triage Algorithm. U.S. HHS CHEMM. <https://chemm.hhs.gov/incident-primer/triage/start-triage>

[5] SALT Mass Casualty Triage Algorithm. U.S. HHS CHEMM. <https://www.chemm.hhs.gov/incident-primer/triage/salt-triage>

[6] Validation of the Interagency Integrated Triage Tool in Papua New Guinea. The Lancet Regional Health - Western Pacific / PubMed. <https://pubmed.ncbi.nlm.nih.gov/34527985/>

[7] Interagency Integrated Triage: Reference card. WHO / ICRC / MSF.

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